Matlab Exercises 2024-25

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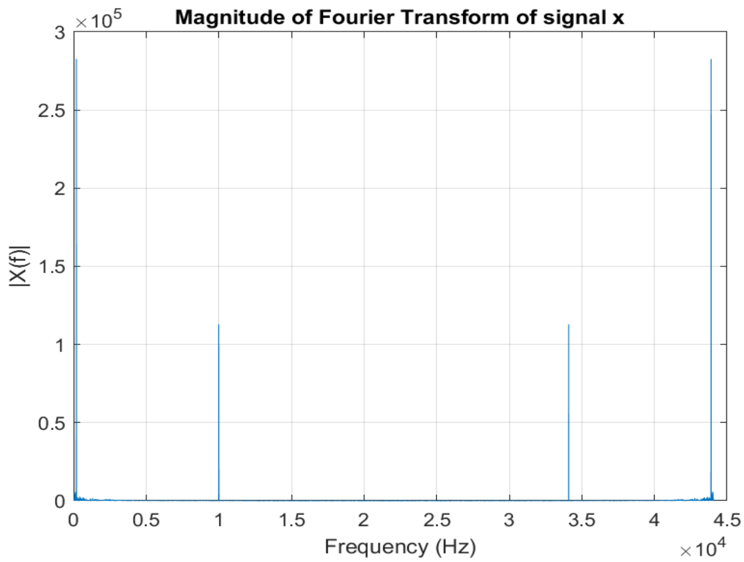
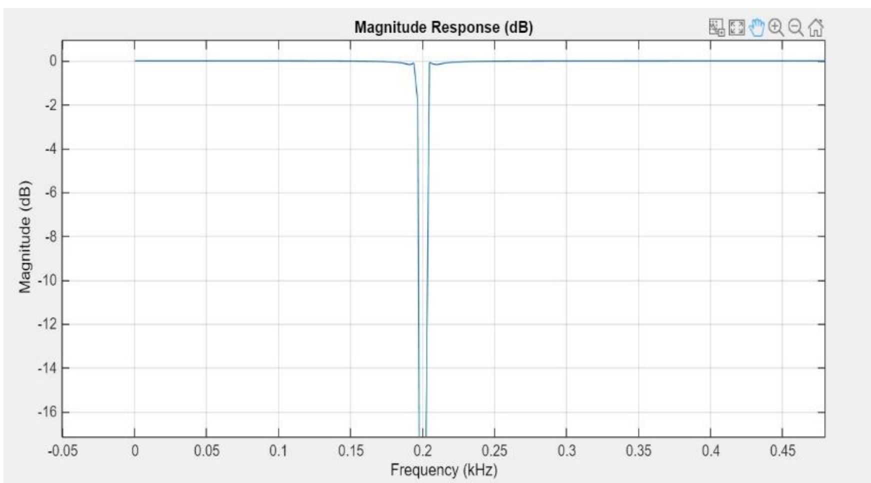
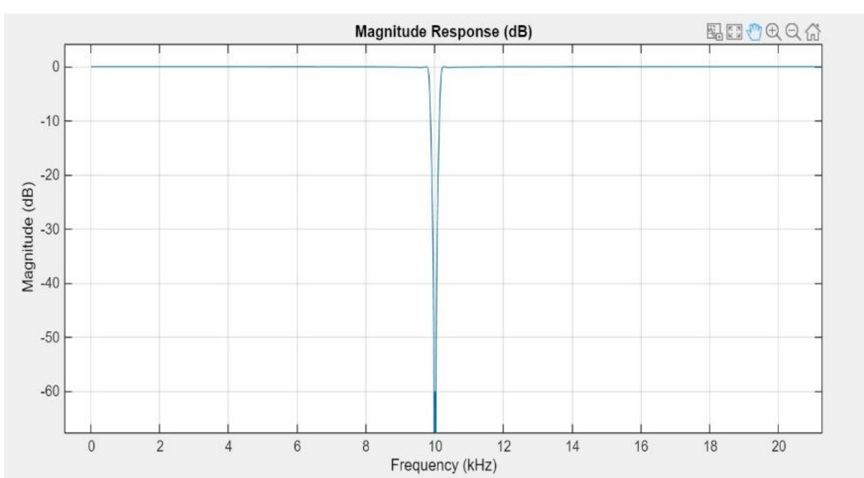
Exercise 1:

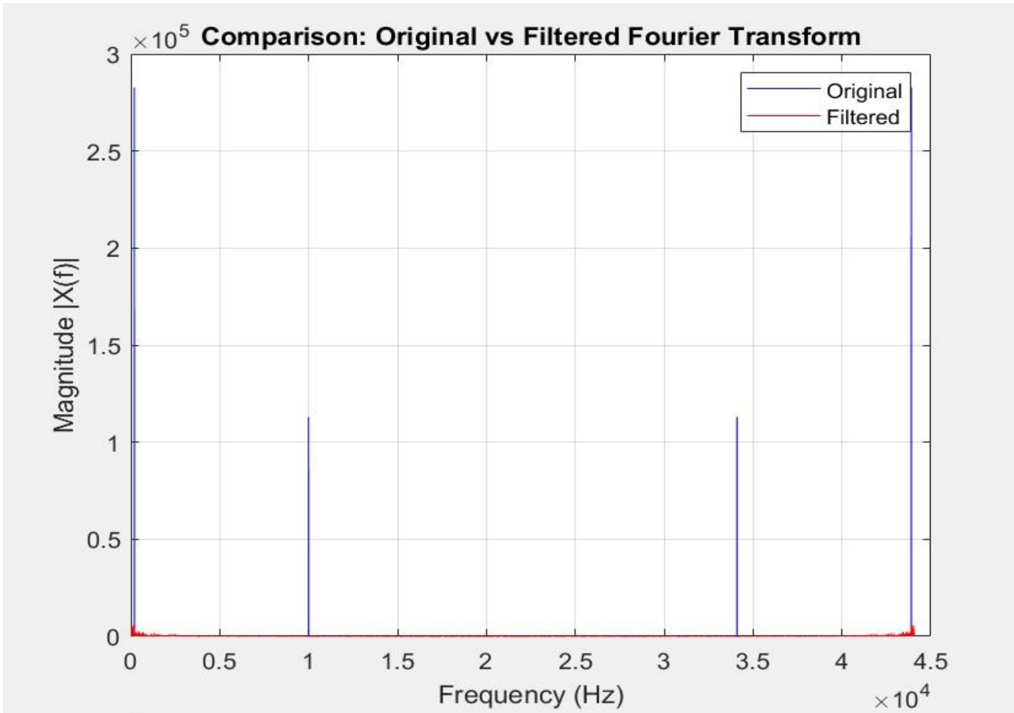
For the first exercise, I used the 2 following notch filters centered at frequencies 200Hz and 10kHz to remove the disturbances. The repeated spikes are therefore also removed by the symmetry of the Fourier transform.







The filters succeeded to remove all the disturbances and the audio file sounded very good. The following graph shows a comparison between the magnitudes of the Fourier transforms of x before and after the filters were applied.

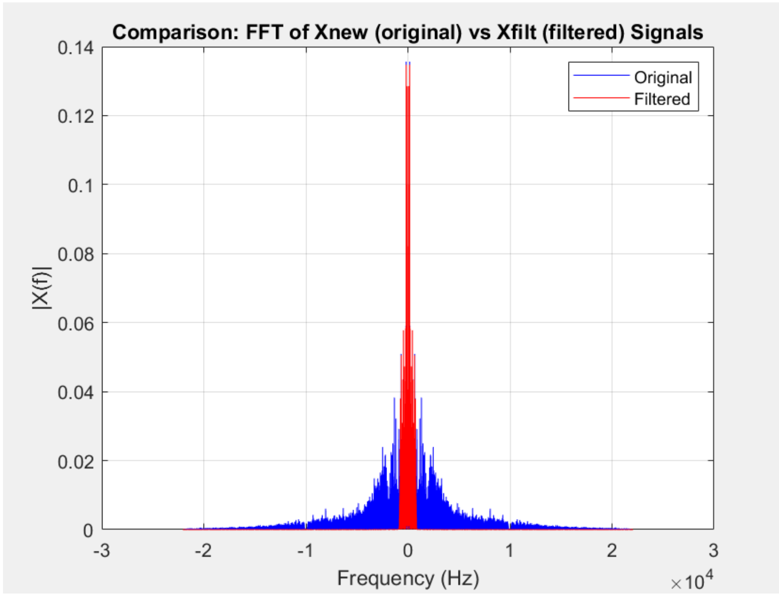


Exercise 2:

After getting the signal x clear from disturbances, we try sampling it at a frequency Fs/25. The produced signal (xsampled) is very muffled and unclear, I believe that is due to aliasing effect since the sampling frequency no longer satisfies Nyquist the condition.

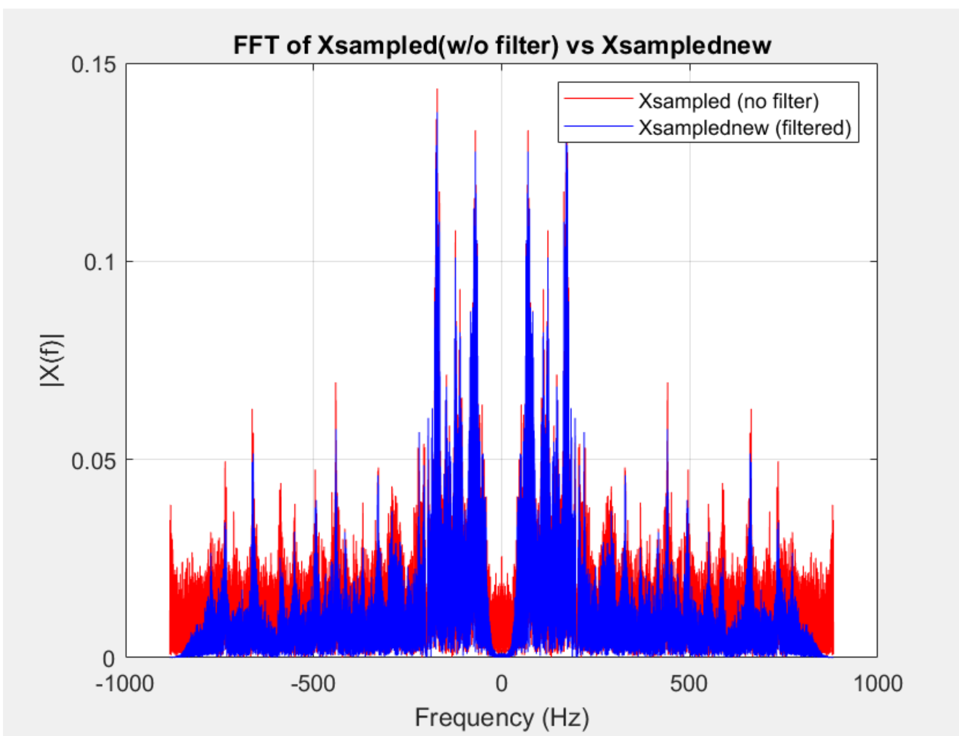
An alternative solution to solve this problem would be to filter the signal x first using a low pass filter that would limit the band of the signal to a frequency that satisfies the Nyquist condition for Fs/25. A suitable cutoff frequency is fstop = Fs/50 since Fs/25 >2 fstop is the Nyquist condtion, and there are no delta functions on fstop so the equality holds.

The following figure shows the comparison between the FFT magnitudes of the original signal x_new and the filtered signal x_filt.



The signal x_filt is then sampled by the new sampling frequency Fs1 = Fs/25, producing the signal xsampled_new. This signal is much clearer than xsampled, which was downsampled without proper filtering. However, it is of course not as clear as x_new since the sampling frequency is much lower, and therefore the sound quality also is.

The following graph shows a comparison between the FFT magnitudes od the signals xsampled and xsampled_new.



It can be seen than the xsampled_new signal is not completely free from aliasing, which could be due to imperfections of the filter used.